Do Macro and Micro RLVR Interventions Compose? A Pre-Registered Factorial Study of Reasoning Breadth and Solution Diversity in a Small Language Model

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***Abstract*—Reinforcement learning with verifiable rewards (RLVR) raises the single-sample accuracy of language models. Whether it also shrinks the set of problems a model can solve is disputed: base models overtake RLVR-trained ones at large k in some studies, longer training expands the boundary in others, and a third line traces the decline to how Pass@k is measured. All of this evidence is automatic, nobody has asked people what RLVR loses, and fixes that act within a rollout group and across prompts have never been tested together. We take up all three gaps on Qwen2.5-0.5B-Instruct trained with GRPO on MATH. A pre-registered 2×3 factorial crosses a macro factor (none, or a topic-level budget on positive advantage) with a micro factor (none, the per-prompt SELF filter, or mean-preserving redistribution over solution modes), with two or three seeds per cell, and a blinded human audit checks what is lost. At 2.0 rollouts per training problem, vanilla GRPO does not shrink the boundary. MATH-500 Pass@1 moves from 30.3% to 30.4%, Pass@32 rises from 73.0% to 75.3%, and every condition gains more problems than it loses. Neither pre-registered contrast is significant: under the budget, swapping the SELF filter for mode redistribution changes the gain from $k = 1$ to $k = 32$ by [-2.7, $+2.4$] points (95% CI), and adding the budget to the redistribution by [-2.1, $+3.6$]. The two annotators agreed on 85% of solution pairs (PABAK $+0.70$) and judged 50 of the 51 agreed pairs to use the same approach. The word-overlap partitioner that splits such pairs into "modes" scores 55% against them, below the 98% of a labeller that always answers "same". The diversity these fixes reweight is mostly wording. Labels of reasoning validity agreed no better than chance, so what was lost stays open. We read the null as a matter of training intensity, not model size. The protocol runs on one 24 GB GPU.**

Code, data and pre-registration: **https://github.com/2005legend/Reasoning-boundary-paradox-of-SLM**

***Index Terms*—reinforcement learning with verifiable rewards, GRPO, Pass@k, reasoning boundary, small language models, credit assignment, human evaluation, pre-registration**

I. Introduction

Group Relative Policy Optimization (GRPO) [1] and related RLVR methods are now the standard way to improve mathematical reasoning in open language models. Whether they teach a model to solve new problems, or only to solve solvable ones more reliably, is disputed. Yue et al. [2] found that base models overtake their RLVR-trained versions when many samples are drawn, a *Pass@k inversion* later analysed as a shrinking reasoning boundary [3]–[6]. Others report the opposite. Prolonged training expands the boundary [7], RLVR comes out ahead when only correct chains of thought are counted [8], and a two-stage account reconciles the two by training duration [9]. A third view blames the measurement: aggregate declines track how few rollouts each training problem receives [10], and Pass@k at large k rewards lucky hits [11]. The question matters most for small language models (SLMs), which depend on sampling-based coverage such as majority voting and verifier reranking.

The evidence on every side is automatic. Pass@k cannot tell a valid solution from a lucky guess. CoT-Pass@k hands that judgment to LLM judges, which accept corrupted reasoning almost as often as clean reasoning [12], and approach-level diversity has been scored by an LLM judge calibrated on 80 human-labelled pairs [13]. As far as we know, no study has asked people what RLVR loses: whether the solutions that disappear are distinct strategies, and whether lost problems were ever solved by valid reasoning.

A third gap concerns the fixes. *Intra-group* methods give rare correct modes of one prompt more credit than common ones [14]–[19], and *cross-prompt* methods reweight whole problems or tasks [3], [10], [20], [21]. The two act on different axes but have not been tested together (ReCo combines two

corrections, both within one prompt [15]). One pre-registered study addresses all three gaps:

- *Is shrinkage real at SLM scale?* A multi-seed measurement on Qwen2.5-0.5B-Instruct [22] trained on MATH [23], with problem-level boundary entries and exits and the number of rollouts per training problem [10].
- *What is lost, according to people?* A blinded audit of (H1) whether our partitioner's solution modes are distinct strategies and (H2) whether the base model's successes on lost problems were valid reasoning.
- *Do the fixes compose?* A 2×3 factorial crossing a topic-level budget on positive advantage (CB, macro) with no micro term, the SELF filter, or intra-group mode-rarity redistribution (MEG).
- The answers: at about two rollouts per training problem RLVR expands the boundary slightly, no fix or combination moves the shrinkage slope by more than about one point per unit $\ln k$, and the modes the fixes rely on are mostly wording.

II. Related Work

Is shrinkage real? Yue et al. [2] showed that base models overtake RLVR-trained ones at large k. Nguyen et al. [3] attribute this to negative interference and winner-take-all reinforcement and propose SELF, which trains only on problems whose greedy answer is wrong. Wu and Xuan [5] argue that RLVR cannot leave the base model's support, Zhou and Li [6] place the contraction at the first decision of a solution, and Zhou [4] diagnoses the inversion. Against this, ProRL [7] reports expansion after prolonged, KL-controlled training, Wen et al. [8] report expansion under CoT-Pass@k, and Yao et al. [9] describe an exploitation stage that shrinks the boundary followed by an exploration stage that expands it. Yuan et al. [10] tie the decline to rollouts per problem and gate problems by success count (BBG), and Dragoi et al. [11] show that Pass@k favours lucky hits. Other work changes the optimizer or the search [24]–[26]. At SLM scale, GRPO dynamics [27], difficulty scaling [28] and reward granularity [29] have been studied, but not shrinkage.

Measuring what is lost. Lee et al. [13] find that embedding similarity tracks phrasing more than strategy and word-bigram overlap tracks approach better, and that 0.7% of GSM8K problems but 33% of MATH problems admit more than one approach. Yang et al. [30] code the strategies of frontier models with AI coders and human adjudication, without RL. Taşaltı et al. [12] find that CoT-Pass@k judges accept corrupted chains, and that Pass@k exceeds CoT-Pass@k by 19.7 points on Qwen2.5-generation solvers. Our audit puts people directly on RLVR outcomes.

Intra-group credit redistribution. Cue-GRPO [14] redistributes positive credit over a partition of the correct rollouts by rarity and restores the mean; we adopt its rule as our micro term. ReCo [15] reweights against concentration at the response and token level, and GCPO [16], DRA-GRPO [17], EDAS [18] and TreeAdv [19] reshape advantages using diversity or structure within the group.

Cross-prompt reweighting. MT-GRPO [21] reweights explicit tasks for worst-task reliability, CurveRL [20] reweights prompts by pass-rate distribution, BBG gates problems by success count [10], and SELF selects by greedy failure [3]. CB works at the level of topics, uses each topic's share of positive advantage as its signal, and keeps state across steps. We ask whether a topic-level and an intra-group intervention compose, not which gate is best.

III. Method

A. GRPO with gated advantages

For a prompt x the policy π_θ samples G completions $y_1, \dots, y_G$ with binary rewards $r_i \in \{0, 1\}$. GRPO [1] uses the group-normalized advantage $\hat{A}_i = (r_i - \mathrm{mean}(r))/(\mathrm{std}(r) + \epsilon)$. Every condition multiplies it by a non-negative gate weight, $\tilde{A}_i = g_i \hat{A}_i$ with $g_i = g_i^{\mathrm{mac}} g_i^{\mathrm{mic}}$, and minimizes the PPO-clip loss [31] averaged over all T completion tokens in the batch [32]:

$$\mathcal{L}(\theta) = -\frac{1}{T} \sum_i \sum_{t=1}^{|y_i|} \min\big(\rho_{i,t} \tilde{A}_i,\ \mathrm{clip}(\rho_{i,t}, 1-\varepsilon, 1+\varepsilon)\, \tilde{A}_i\big), \tag{1}$$

where $\rho_{i,t}$ is the token probability ratio. Updates are on-policy with no KL penalty, and vanilla GRPO is $g_i \equiv 1$. Fig. 1 shows one training step.

B. Macro gate: topic budget (CB)

Topics are the $C = 7$ MATH subjects. After each step CB updates an exponential moving average of each topic's *spend*, the mean positive advantage of its rollouts in that step: $s_c \leftarrow (1-\eta)\, s_c + \eta\ \mathrm{mean}_{i:c(x_i)=c} \max(\hat{A}_i, 0)$ with $\eta = 0.05$. Negative advantage never uses up a topic's budget. With the overspend ratio $\rho_c = s_c / \frac{1}{C} \sum_{c'} s_{c'}$,

$$\tilde{g}_c = \delta^{\max(0,\ \rho_c - \vartheta)}, \qquad \vartheta = 1.1,\ \delta = 0.18, \tag{2}$$

so a topic at $1.5\times$ the average spend gets half the credit of one within budget. The gate acts only on rollouts with positive advantage; their weights are clipped to $[0.3, 3]$ and rescaled to mean 1 over the step, so CB moves credit between topics instead of lowering the learning rate. The original specification ($\vartheta = 1.5$, $\delta = 0.98$, down-weighting only) never fired in the pilot, where spend ratios stayed within $[0.38, 1.46]$, so we changed it before any main run and recorded the change in the pre-registration.

C. Micro gates: SELF filter and mode redistribution (MEG)

SELF. Following [3], a prompt whose greedy answer is already correct gets weight $\lambda = 0$ and every other prompt weight 1. The greedy answer is decoded as one extra row of the same generation call. We implement the selection rule only, not SELF's forward-KL term.

MEG. Let $\mathcal{C}$ be the N correct rollouts of a group, partitioned into solution modes, and let $|M(i)|$ be the size of the mode containing rollout i. Following Cue-GRPO [14],

$$w_i = \frac{N\, |M(i)|^{-\alpha}}{\sum_{j \in \mathcal{C}} |M(j)|^{-\alpha}} \quad (i \in \mathcal{C}), \qquad w_i = 1 \quad (i \notin \mathcal{C}), \tag{3}$$

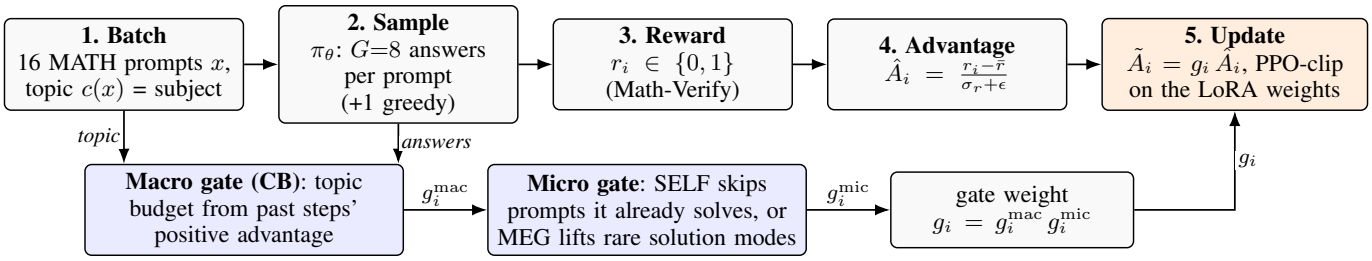


Fig. 1. One training step of gated GRPO. The macro gate knows each prompt's topic and how much positive advantage that topic has absorbed in past steps; the micro gate sees the current group's answers (and greedy answer). Their product scales the advantage. Vanilla GRPO sets both gates to 1.

TABLE I
PARTITIONER CALIBRATION ON BASE-MODEL ROLLOUTS (GROUPS WITH ≥ 4 CORRECT). $\star$ = SELECTED.

MiniLM cosine τ	0.80	0.85	0.90	0.95	
median modes	1.0	1.0	1.0	3.5	
multi-mode groups (%)	2	12	48	88	
Word-bigram threshold	0.30	0.40	0.50	0.60	0.70
median modes	5.0	3.0$\star$	2.0	1.0	1.0
multi-mode groups (%)	100	90	60	35	5

TABLE II
THE 2×3 FACTORIAL. COMPOSITES MULTIPLY THE MACRO AND MICRO GATES.

macro \ micro	none	SELF filter	MEG redistribution
none	VANILLA	SELF	MEG
CB topic budget	CB	CB$\times$SELF	CB$\times$MEG (HEG-GRPO)

with $\alpha = 0.8$, clipped to $[0.3, 3]$ and rescaled to mean 1 over $\mathcal{C}$. MEG changes *which* correct solutions get credit, not how much; an earlier down-weight-only version was confounded with a lower learning rate.

Mode partitioner. Modes come from agglomerative clustering of correct completions, on MiniLM [33] sentence embeddings [34] or on word-bigram Jaccard distance. A pre-registered rule, applied to base-model rollouts before training, picks the candidate whose median number of modes per group lies in $[2, 4]$ and is closest to 3. It selected word-bigram Jaccard, threshold 0.4 (median 3.0 modes, 90% of groups with more than one; Table I). The embedding candidates put almost every group's correct rollouts into a single mode, which would make MEG a no-op.

IV. EXPERIMENTAL SETUP

Training. We train Qwen2.5-0.5B-Instruct [22] with LoRA [35] adapters on the MATH training split [23] (7,496 problems with a boxed answer) under the standard "reason step by step and put the final answer in `\boxed{}`" prompt. Table II lists the six conditions and Table III every setting. The reward is binary correctness checked with Math-Verify [36], with a guarded symbolic fallback. The budget is 0.26 epochs, or 2.0 rollouts per training problem. Seed s fixes the data order and LoRA initialization across conditions, so conditions are compared in seed-paired blocks.

TABLE III
TRAINING AND EVALUATION SETTINGS, THE SAME FOR ALL CONDITIONS.

Model	Qwen2.5-0.5B-Instruct, bf16; LoRA r=16, α=32, all projections
Optimizer	AdamW, lr 1×10^{-5}, cosine, 10% warm-up, grad-norm clip 1.0
Batch, steps	16 prompts $\times$ G=8 rollouts; 120 steps; MATH levels 1–5
Sampling	T=0.7, top-p 0.95, $\leq$1,024 new tokens
Loss	PPO-clip ε=0.2, on-policy, token-level mean, no KL
CB / SELF / MEG	η=0.05, ϑ=1.1, δ=0.18 / λ=0 / α=0.8
Evaluation	MATH-500, n=32 samples, T=0.6, top-p 0.95
Hardware	NVIDIA A10G 24 GB (AWS g5.xlarge), one run per GPU

Evaluation and outcome. Every model, including the base model, answers all of MATH-500 [37] 32 times per problem. Pass@k for $k \in \{1, 2, 4, 8, 16, 32\}$ uses the unbiased estimator [38]. MATH-500 shares no problem statement with the training split. The pre-registered primary outcome is the *shrinkage slope*, the least-squares slope of $\Delta(k) = \text{Pass@}k(\text{run}) - \text{Pass@}k(\text{base})$ against $\ln k$ in percentage points per unit $\ln k$; a negative slope means gains at small k turn into losses at large k. We also count boundary *exits* (problems the base model solves at least once in 32 samples and the trained model never does) and *entries* (the reverse).

Pre-registered analysis. Contrasts are linear combinations of condition slopes, paired by seed. The primary contrasts are P1, CB$\times$MEG $-$ CB$\times$SELF (redistribution vs. filtering under the budget), and P2, CB$\times$MEG $-$ MEG (does the macro level add to the micro level?). Secondary contrasts are the two macro$\times$micro interactions (S1, S2), reported as bounds, and the macro main effect (S3); E1 is exploratory. Each contrast has a hierarchical bootstrap interval over seeds and problems (2,000 resamples), and an effect is claimed only if its Holm-adjusted [39] $p < 0.05$ and its interval excludes zero. The pre-registration also fixed how outcomes are worded, including that a vanilla run without shrinkage is reported as such.

Human audit (pre-registered, secondary). Two annotators who are not authors, Shruti K (BBA) and Sri Prathin (final-year B.E. CSE student), label two blinded sheets without seeing condition, group or partitioner output. *H1:* 60 pairs of correct base-model solutions to the same training problem, half placed in one mode by the partitioner and half in two,

TABLE IV
MATH-500 PASS@k (%, MEAN OVER SEEDS), SHRINKAGE SLOPE (PP PER UNIT $\ln k$; HIGHER IS BETTER) AND BOUNDARY ENTRIES/EXITS (OF 500 PROBLEMS).

Condition	P@1	P@4	P@16	P@32	Slope	In / Out
Base model	30.3	50.1	66.3	73.0	–	–
VANILLA	30.4	51.1	68.4	75.3	+0.66	36.3 / 25.0
SELF	30.8	51.1	68.0	75.2	+0.48	36.5 / 25.5
MEG	30.4	50.8	67.1	73.5	+0.13	30.7 / 28.3
CB	31.2	51.3	67.9	74.6	+0.21	34.0 / 26.0
CB×SELF	31.4	51.6	68.5	75.3	+0.37	32.3 / 20.7
CB×MEG	30.8	51.3	68.1	74.7	+0.35	33.3 / 25.0

TABLE V
CONTRASTS ON THE MATH-500 SHRINKAGE SLOPE (PP PER UNIT $\ln k$), PAIRED BY SEED, WITH HIERARCHICAL BOOTSTRAP CIS; THE LAST COLUMN RESTATES THE CI AS POINTS OF GAIN FROM k=1 TO k=32. p: HOLM-ADJUSTED FOR P1–P2.

Contrast	Est.	95% CI	p	CI, k=1→32
P1 CB×MEG − CB×SELF	−0.02	[−0.79, +0.68]	1.000	[−2.7, +2.4]
P2 CB×MEG − MEG	+0.22	[−0.60, +1.05]	1.000	[−2.1, +3.6]
S1 CB×MEG interaction	+0.38	[−0.94, +1.69]	0.493	[−3.2, +5.9]
S2 CB×SELF interaction	+0.46	[−0.71, +1.68]	0.396	[−2.5, +5.8]
S3 CB main effect	−0.18	[−0.59, +0.25]	0.360	[−2.0, +0.9]
E1 CB×MEG − Vanilla	−0.31	[−0.91, +0.30]	0.291	[−3.2, +1.1]

labelled SAME or DIFFERENT approach under the definition of [13]; modes count as approach-level if the partitioner's balanced accuracy against the consensus reaches 70%. *H2:* MATH-500 problems the base model solves but vanilla GRPO never solves in a majority of seeds (*exited*), each matched to a retained problem with the closest base solve count. Annotators judge one correct base-model solution per problem VALID or FLAWED, and the groups are compared with a two-sided Fisher exact test.

Reproducibility. Code, logs, annotation sheets and the script behind every number are in https://github.com/2005legend/Reasoning-boundary-paradox-of-SLM. The pre-registration was written on 2026-10-03 and changed only by dated entries; it was first committed after the runs (`4cdc43a`), so its dates are self-reported.

V. RESULTS

A. Shrinkage at 0.5B

Vanilla GRPO does not reproduce the inversion at this budget. Pass@1 moves from 30.3% to 30.4% and Pass@32 rises from 73.0% to 75.3% (Table IV). The curve stays above zero up to $k = 32$, with slope $+0.66$ (Fig. 2, left). On average 36.3 problems enter the boundary and 25.0 leave it, and every other condition also has a positive slope and more entries than exits. Each training problem received 2.0 rollouts on average, the low-visit end that SLM budgets allow; aggregate shrinkage has been tied to how often a problem is revisited [10].

B. Pre-registered contrasts

All six slopes are positive, from $+0.13$ (MEG) to $+0.66$ (VANILLA), and no gate beats vanilla. Neither primary contrast clears the pre-registered bar (Table V). In points of gain between $k = 1$ and $k = 32$, P1 lies within $[-2.7, +2.4]$ and P2 within $[-2.1, +3.6]$, so any difference between the micro terms is under about three points. The interactions are wider, $[-3.2, +5.9]$ and $[-2.5, +5.8]$, which admits effects larger than vanilla's whole gain at $k = 32$; with two seeds per cell the interaction question stays open. Both primary 90% intervals lie inside ± 1.0 pp per $\ln k$ (± 3.5 points at $k = 32$), so we call P1 and P2 equivalent to zero within that bound, which we chose after seeing the data; the interactions do not meet it. With the between-seed spread we observed, three seeds detect a difference of about 1.4 (P1) and 2.0 (P2) pp per $\ln k$ at 80% power, and five seeds would detect 0.7 and 1.0. We read the primary contrasts as a bounded null, as pre-registered. The gates also made runs less reproducible: the slope's standard deviation across seeds is 0.04 for VANILLA and 0.40 for MEG (0.25 and 0.28 for the composites), a descriptive comparison at three seeds.

C. What each level does

Mode entropy of the correct rollouts starts at 0.63 and does not decay under vanilla GRPO (Fig. 2, right). At evaluation, vanilla's correct answers to a problem fall into 7.39 modes on average, against 6.73 for the base model, and its largest mode holds 39% of them against 43%. There is no collapse for MEG to undo. SELF skips the 59% of prompts it already solves, for a mean positive weight of 0.29. Changes are uneven across subjects (Fig. 3): under vanilla GRPO, Pass@32 rises by – pp in Counting & Probability and by – pp in Geometry. The macro gate binds but changes nothing. In the CB runs a subject's share of positive credit does not predict its Pass@32 change (Spearman $\rho = +0.01$, $n = 28$ subject-runs), which leaves the premise of a topic budget unsupported.

Boundary exits barely separate the conditions. CB×SELF loses the fewest problems (20.7) and MEG the most (28.3), against 25.0 for VANILLA. Across the 16 runs, those that gained more distinct correct modes lost slightly more problems (Spearman $\rho = +0.57$, $p = 0.02$), but this does not survive correction over the 7 correlations we computed (Benjamini–Hochberg $p = 0.16$). Since the next subsection shows these modes are mostly wording, we treat it as a hypothesis.

D. What is lost: the human audit

H1, are the modes approaches? The annotators called 58 and 51 of the 60 pairs the same approach, although the partitioner had split half of them. They agreed on 51 pairs (85%), 50 SAME and 1 DIFFERENT. Cohen's κ is only 0.13 because almost every label falls in one class, which depresses κ however well raters agree; the prevalence-adjusted PABAK is $+0.70$. Against the consensus the partitioner scores 55%, below the 98% of a labeller that always says SAME. The pre-registered balanced accuracy is 77%, formally above the 70% bar, but one of its two classes holds a single pair, so we do not rely on it. We read the modes as surface clusters, and MEG as redistributing credit over them.

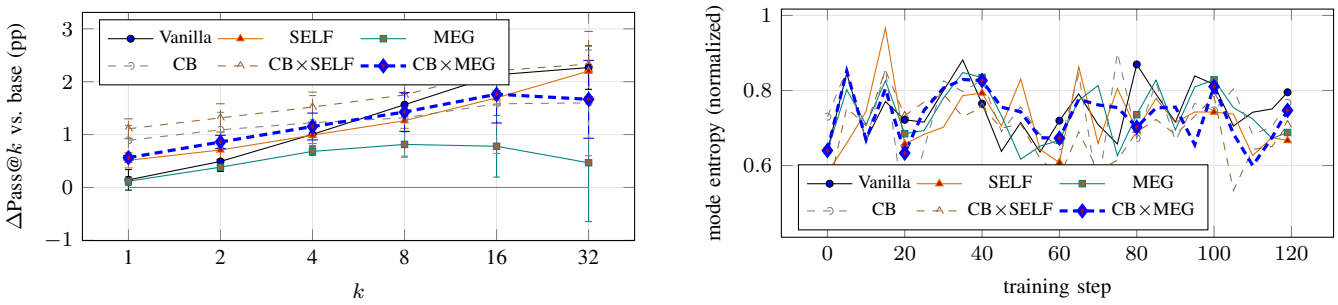


Fig. 2. Left: change in MATH-500 Pass@k relative to the base model (mean ± s.d. over seeds); a curve that fell below zero at large k would mean a shrunk boundary. Right: normalized entropy of the solution-mode distribution among correct rollouts during training (mean over seeds).

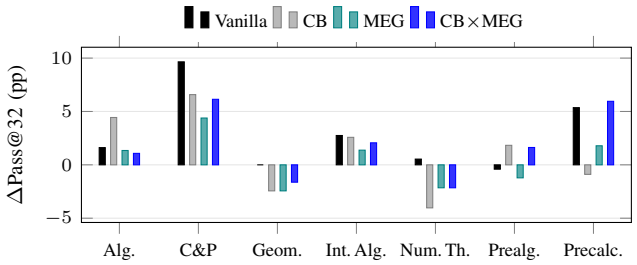


Fig. 3. Change in MATH-500 Pass@32 relative to the base model by subject, the macro gate's topics (mean over seeds).

H2, what is lost? Despite its net gains, vanilla GRPO lost 20 problems in a majority of seeds. The annotators disagreed on whether the base model's successes on them were valid: A rated 11 of 40 solutions valid and B 28 ($\kappa = -0.06$, agreement on 15 items). On those items, 83% of exited solutions were valid (CI [44, 97], $n = 6$) against 22% of retained ones (CI [6, 55], $n = 9$; Fisher $p = 0.041$). Read literally, the pre-registered rule calls this a genuine loss, but a subset chosen by agreement at chance-level reliability cannot carry that; each annotator alone shows the same direction without significance (A $p = 0.480$, B $p = 0.082$). We report H2 as inconclusive.

VI. Discussion and Limitations

Training intensity, not size. With about two rollouts per training problem and a policy that moved little (final KL to the base model about ten times its bf16 noise floor), RLVR on a 0.5B model expands the boundary slightly. Accounts that tie shrinkage to repeated reinforcement of saturated problems [10] or to a long exploitation phase [9] predict exactly this, and the reported inversions [2], [3] come from far longer training. We read our null as a statement about the regime a single-GPU budget reaches, not about model size. It suggests a direct test: raise rollouts per problem at 0.5B and shrinkage should appear. We have registered that test: five seeds per cell and 240-step runs scored at steps 120, 180 and 240. Our prediction, made before running it, is that vanilla's slope declines over those steps; if it does not, the intensity reading fails. **The partition is the weak link.** Embedding partitions put nearly every group in one mode (Table I), and the bigram partition finds about three modes that people see as one approach. Neither tracks what intra-group methods assume it tracks, which matters for any method that reweights over such partitions [14]–[18]. This fits Lee et al.'s finding that a third of MATH problems admit more than one approach [13]: the alternatives exist in the dataset but rarely appear in what a 0.5B model samples. We do not claim that shrinkage is absent at larger budgets; the bounds in Table V are what a longer study would need to beat.

Limitations. One model at one size, LoRA, one RL algorithm and short training (120 steps, cut from a planned 240), evaluated only on MATH-500. Two or three seeds per cell leave the interactions open, and each run has one full evaluation, so we show no trajectory over training. SELF omits its forward-KL term, and no random-partition control was run. The annotators differed in strictness on H2, and the pre-registered third annotator was not used. The correlations above are descriptive and were added after the pre-registration.

VII. Conclusion

In a pre-registered study of Qwen2.5-0.5B-Instruct with a blinded human audit, shrinkage does not appear at about two rollouts per training problem, and neither a topic-level budget, intra-group mode redistribution, nor their combination moves the shrinkage slope by more than about one point per unit $\ln k$. People judged the partitioner's modes to be wording variants of one approach. The protocol runs on one 24 GB GPU in 86 GPU-hours and is released.

Acknowledgment

We thank the two annotators. This work used AWS credits awarded to our team for placing among the top 500 teams in the Amazon ML Challenge 2026; early experiments ran on a local NVIDIA RTX 3060 (12 GB). AWS compute: 86 A10G GPU-hours, about $87. An AI assistant (Claude, Anthropic) was used to write analysis code and to draft and edit parts of the text; the authors checked the results against the logs and take responsibility for the content.

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